

BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `~/Desktop/4750 Cloned Repositories/hw3-wrn25-hw3`

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day⁻¹, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day⁻¹, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance d km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m ³ /d	10,000 m ³ /d	15,000 m ³ /d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

→

For the equation:

$$\bullet U \frac{dC}{dx} = k_a(C_s - C) - k_c B_0 e^{-\frac{k_c x}{U}} - k_n N_0 e^{-\frac{k_n x}{U}}$$

The Streeter-Phelps Solution is below:

$$\bullet C(x) = C_s(1 - \alpha_1) + C_0\alpha_1 - B_0\alpha_2 - N_0\alpha_3$$

With:

$$\begin{aligned} \bullet \alpha_1 &= e^{-\frac{k_a x}{U}}, \\ \bullet \alpha_2 &= \left(\frac{k_c}{k_a - k_c}\right) \left[e^{-\frac{k_c x}{U}} - e^{-\frac{k_a x}{U}}\right] \end{aligned}$$

- $\alpha_3 = \left(\frac{k_n}{k_a - k_n}\right) \left[e^{\frac{-k_n x}{U}} - e^{\frac{-k_a x}{U}} \right]$

We are given values for the reaeration rate, k_a , and the decay rates of CBOD and NBOD, k_c and k_n . So we can calculate this equation for $C(x)$ directly.

-> The graph of this analytic solution is below:

```

Cs = 10
C01 = 7.273
C02 = 4.897
B01 = 9.091
B02 = 8.735
N01 = 7.723
N02 = 7.838
ka = 0.55
kc = 0.35
kn = 0.25
U = 6

x1 = range(0, 15, length=1000)
x2 = range(15, 50, length=1000)
a1 = exp.(-ka*x1/U)
a2 = (kc/(ka-kc))*(exp.(-kc*x1/U) - exp.(-ka*x1/U))
a3 = (kn/(ka-kn))*(exp.(-kn*x1/U) - exp.(-ka*x1/U))

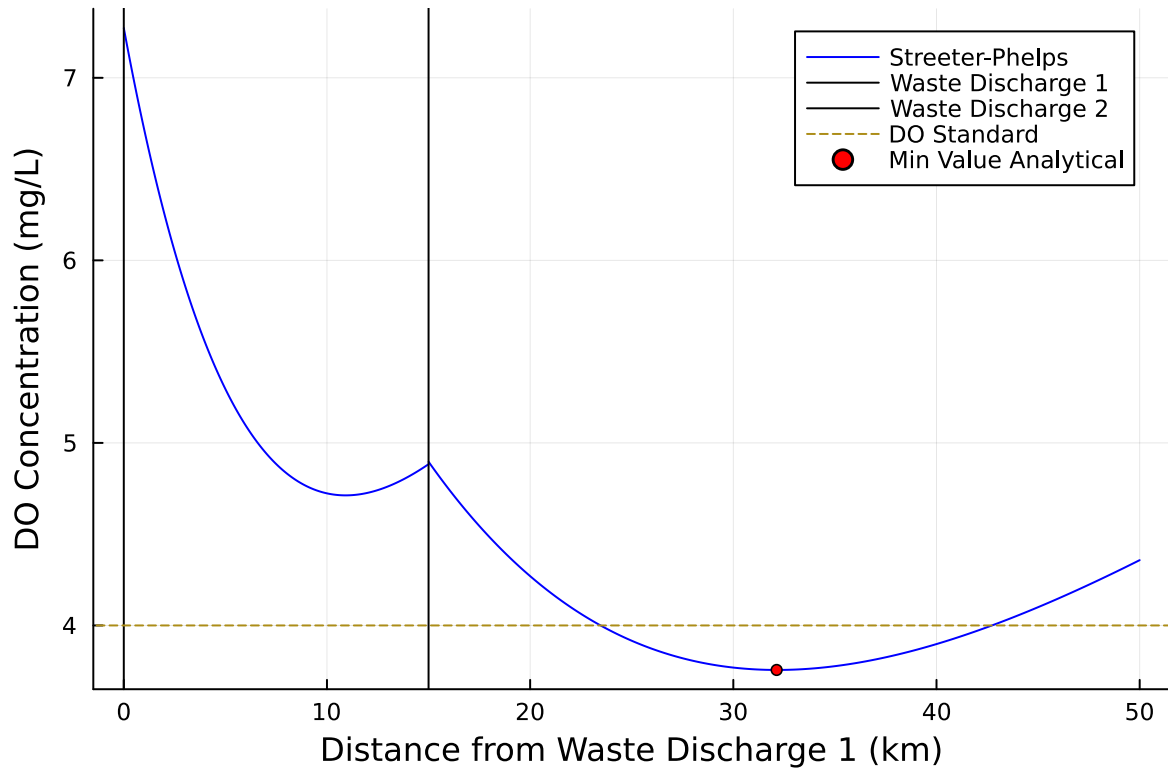
C1 = Cs*(1 .- a1) + C01*a1 - B01*a2 - N01*a3
C2 = Cs*(1 .- a1) + C02*a1 - B02*a2 - N02*a3

p1 = plot(x1,C1, xlabel = "Distance from Waste Discharge 1 (km)", ylabel = "DO Concentration",
label = "", color =:blue)
plot!(p1, x2, C2, label = "Streeter-Phelps", color =:blue)
vline!([0], label = "Waste Discharge 1", color =:black)
vline!([15], label = "Waste Discharge 2", color =:black)
hline!([4], label = "DO Standard", linestyle=:dash)
scatter!(p1, [x2[490]], [3.7562], marker =:circle, markersize =3, color =:red, label = "Min V

#Code to find the minimum value of DO
#min_value = 8.0
#x_value = 0
#for i in 1:1000
    #if C2[i] < min_value
        #min_value = C2[i]
        #x_value = i

```

```
#end
#end
#print(x_value)
```



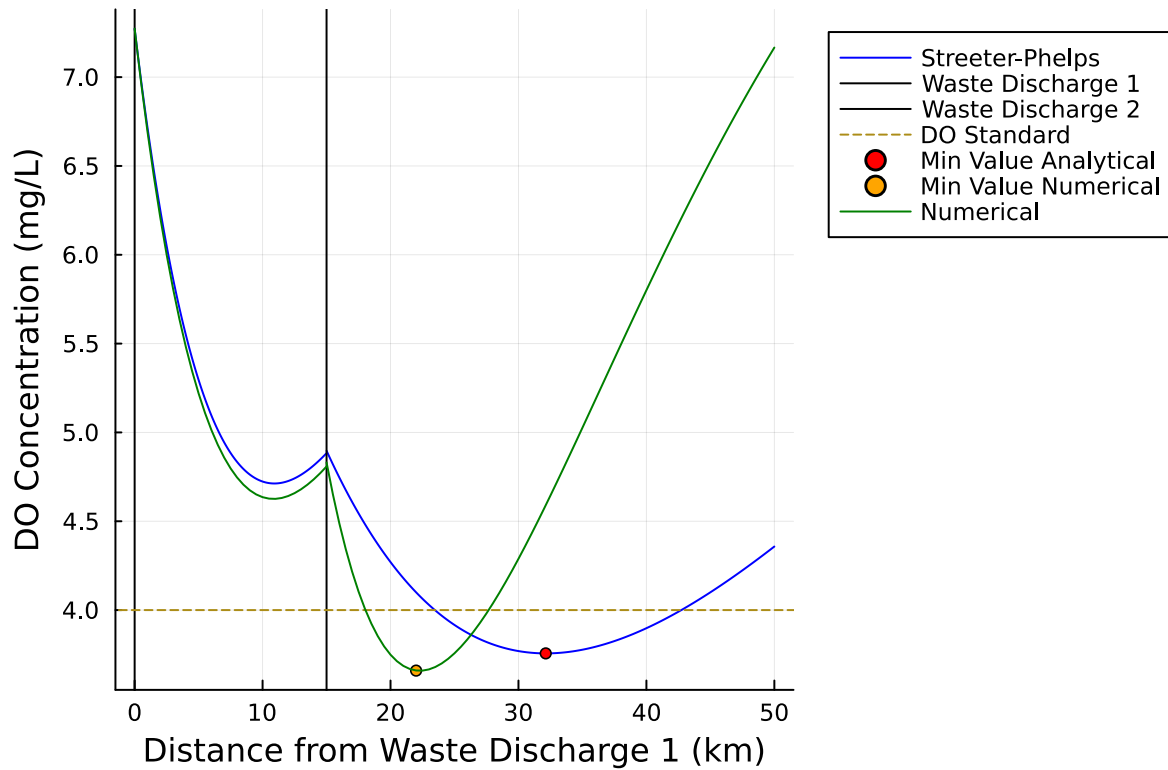
-> The figure above shows the dissolved oxygen concentrations in mg/L in the river flow as it moves from 0km to 50km away from the first waste discharge. The second waste discharge is located at 15km. The value of the minimum level of DO is 3.756mg/L at about 32.15km.

-> Code Explanation: I added the governing equations for each alpha variable into my code and had them depend on a large range of x values. I split the x values up into two ranges, for before and after the second waste discharge. To find values for N_0 and B_0 for each discharge condition I used the volumetric concentration equation and the available information in the table, and the values at $x = 15\text{km}$ to solve for each value. I added vertical lines using the `vline!()` function to show clearly where the waste discharges were located. To find the minimum value of the plotted points I used a for loop to iterate over each value and keep the lowest it found (3.756mg/L).

Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

```
function do_numerical(L, Δx, C0, B0, N0, ka, kn, kc, Cs, U)
    n = Int(L / Δx)
    C = zeros(n + 1)
    B = zeros(n + 1)
    N = zeros(n + 1)
    C[1] = C0
    B[1] = B0
    N[1] = N0
    for i = 1:n
        B[i+1] = B[i] * exp(-kc * Δx / U)
        N[i+1] = N[i] * exp(-kn * Δx / U)
        C[i+1] = C[i] + (Δx / U) * (ka * (Cs - C[i]) - kc * B[i] - kn * N[i])
    end
    return (C, B, N)
end
C1_, B1_, N1_ = do_numerical(15, 0.5, 7.273, 9.091, 7.723, .55, .25, .35, 10, 6)
C2_, B2_, N2_ = do_numerical(35, 0.5, 4.831, 8.735, 7.834, .55, .25, .35, 10, 6)
scatter!(p1, [22], [3.6593], marker=:circle, markersize=3, color=:orange, label = "Min Va
plot!(p1, 0:0.5:15, C1_, label = "Numerical", color=:green, legend=:outertopright)
plot!(p1, 15:0.5:50, C2_, label = "", color=:green)
display(p1)
```



→ The figure above shows the analytical solution (Streeter-Phelps) and the numerical approximation with $\Delta x = 0.5\text{km}$ for the dissolved oxygen levels in the river at increasing distance away from the first Waste Discharge. We see that both solutions dip below the DO standard of 4mg/L , and the numerical solution reaches its minimum much earlier.

→ Code Explanation: I used code seen in lecture to model the numerical and adapted it for this problem and with accurate values for variables. I used the `plot!()` function to add the new plots generated from the `do_numerical()` function to the same plot from problem 1.1. I used the `scatter!()` function to add the second minimum value point. To find the initial conditions for the second part of the numerical graph (after 15km), I printed out the important values at $x = 15\text{km}$ then solved the volumetric concentration equations to find the slightly altered values for initial conditions from 15km to 50km .

→ The min value has changed to 3.6593mg/L at approximately $x = 22\text{km}$. This minimum is less than what we got in problem 1.1, and it occurs earlier. This is because when we approximate the derivative while a function is decreasing and concave up, we end up overestimating the overall magnitude of the rate of decrease. We can see this in the first section (0km to 15km) where the minimum ends up being slightly less than the analytical solution. This means that we have a smaller value for initial concentration at 15km after the second discharge begins mixing with the flow. In addition, the values for B_0 and N_0 for after Waste discharge 2 are slightly different than they were in the analytical solution. All of this leads

to a quicker decrease of DO, and eventually a faster recovery since the reaeration rate also increases faster as it depends on the difference between the saturation concentration and the current concentration.

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

References

List any external references consulted, including classmates.